Relax and stretch



Introduction

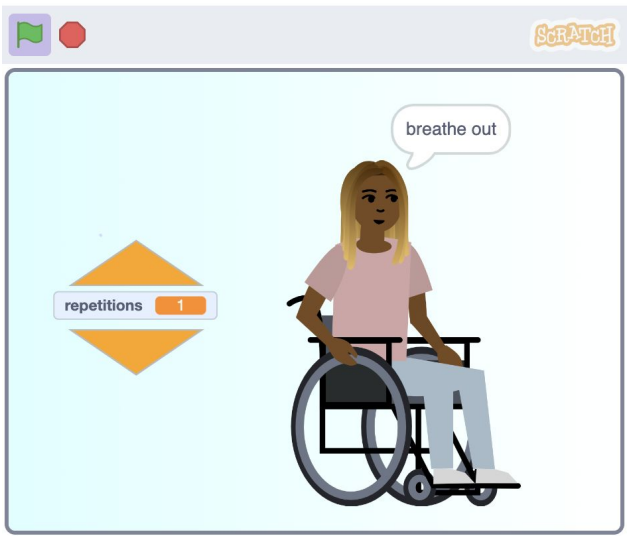
What you will make

You will create a program to demonstrate some mindful, relaxing exercises that you can carry out while sitting down, and even add a new exercise of your own.

- Click on the green flag to start the exercise routines.
- Click on the orange **up** or **down** arrows to change the number of times that each exercise repeats

You will learn to -

- use a **condition** to control a **variable**
- use a **variable** to control the number of **repetitions**
- use the **Text to Speech** extension blocks

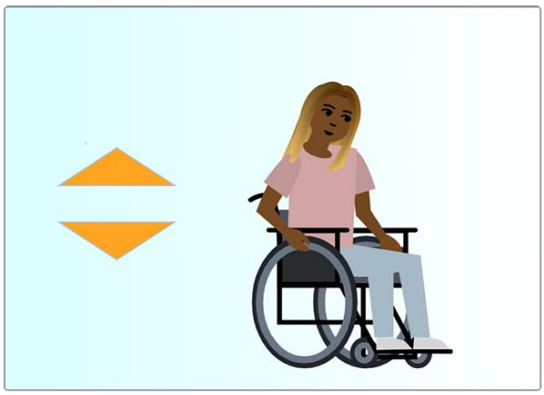


How will you open the project

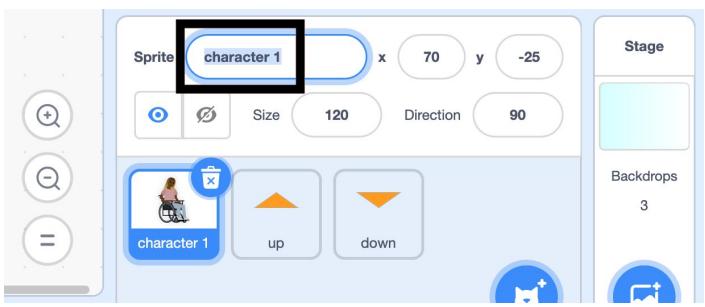
- **Online:** Open the **starter project** in Scratch here - rpf.io/relax-stretch-on
- **Offline:** Open the **project starter file** in the Scratch offline editor here - rpf.io/p/en/relax-stretch-go . If you need to, you can **download and install Scratch here** - https://scratch.mit.edu/download .

1. Prepare your character

Step 1:
First, open the starter project.
You should see a character sprite and two arrows: up and down.

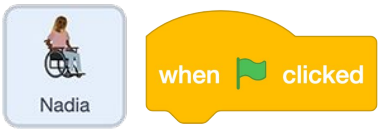


Step 2:
Select the **character 1** sprite.
Click on the **dialogue box** that shows the sprite's name. We have named this sprite **Nadia**. Replace character 1 with the name of your choice.

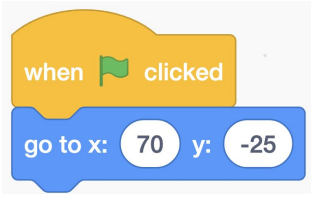


Next, you will give the **Nadia** sprite setup blocks such as `position` on the Stage.

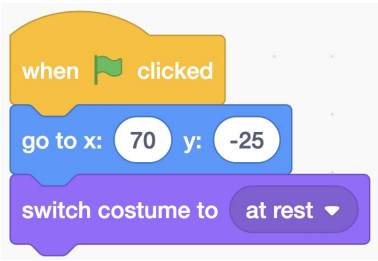
Step 3:
With the **Nadia** sprite still selected, add a `when green flag clicked` block.



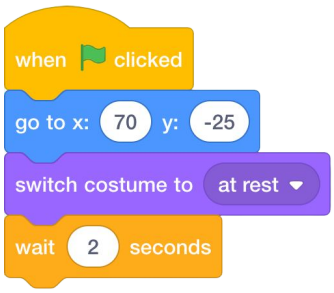
Step 4:
Below that, add a `go to x: y:` block with `x` set to `70` and `y` set to `-25`.



Step 5:
Add a `switch costume to` block, and set the costume to `at rest`.



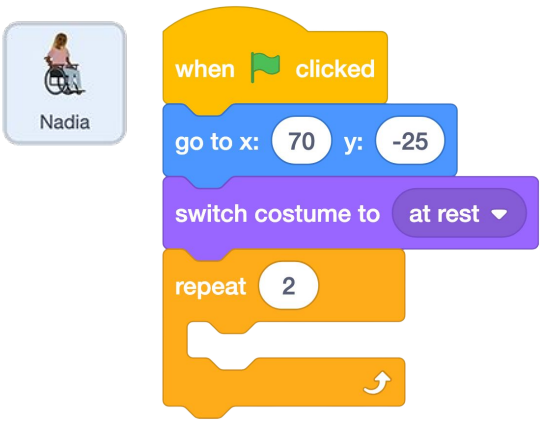
Step 6:
Finally, add a `wait` block and set it to `2` seconds so that the first exercise doesn't begin too suddenly:



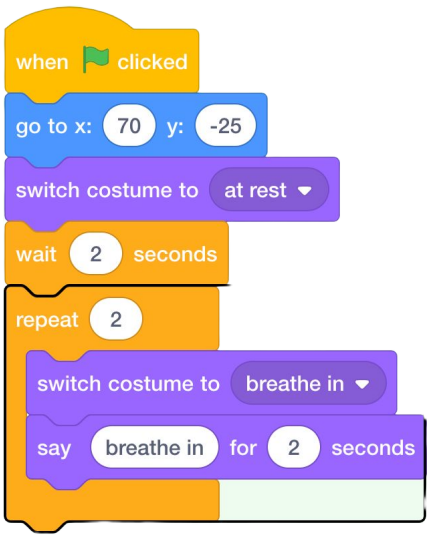
2. Add a breathing exercise

Now, you will start to code a relaxing breathing exercise using different sprite costumes.

Step 1:
Select the **Nadia** sprite and add a `repeat` loop to the existing code. Set the number of `repeats` to `2`.



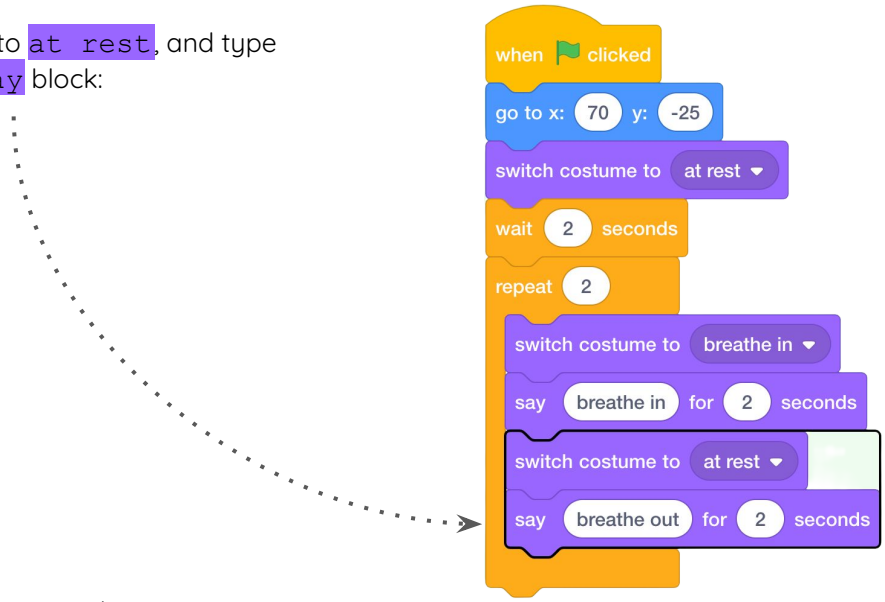
Step 2:
Inside the `repeat` loop, add a `switch costume to` block and a `say` block.
Set the costume to `breathe in` and set the `say` text to `breathe in`:



The `2` in `say breathe in for 2 seconds` tells Scratch how long to show the speech bubble on the Stage.

Now, within the `repeat` loop, add another costume change with some accompanying text.

Step 3:
Set `switch costume` to `at rest`, and type `breathe out` in the `say` block:



You have finished the first exercise!

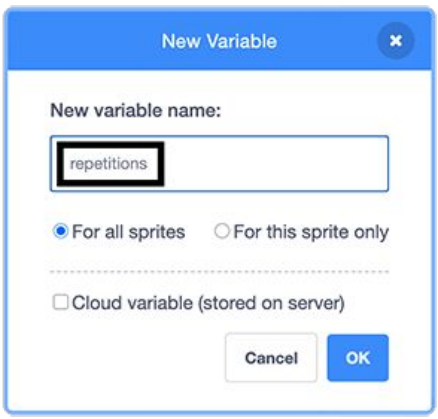
To test your script so far, click on the green flag.

- Did the costume change to look like the sprite was breathing?
- Did your program repeat the routine twice?

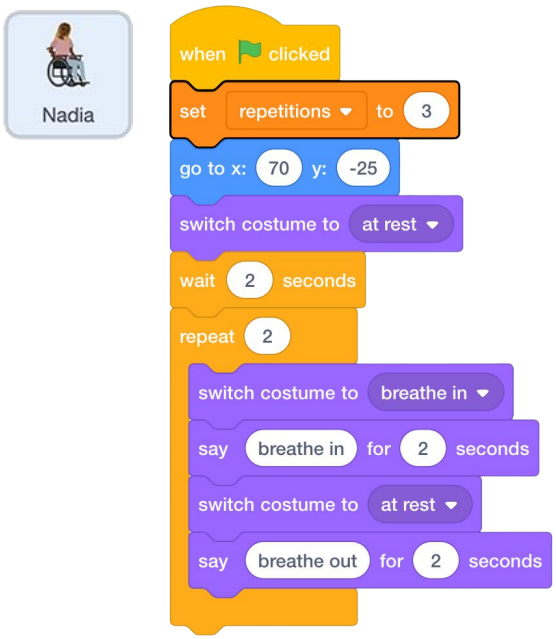
3. Control the repetitions

Now, you will create a variable that will eventually allow a user to control the number of repetitions of each exercise.

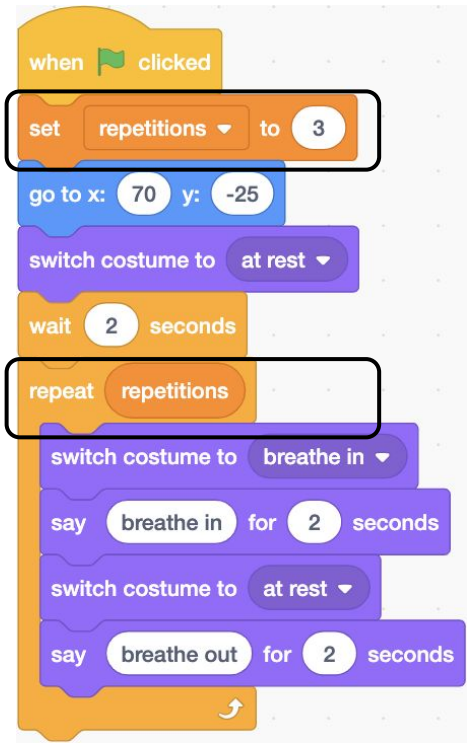
Step 1:
Go to the `Variables` blocks menu, click on **Make a Variable**, and name it `repetitions`.



Step 2:
With the **Nadia** sprite selected, add a `set repetitions to` block and set the value to `3`



Step 3:
Next, you will use repetitions to control the number of times that the exercise repeats.
Add a rounded `repetitions` block as the value of the `repeat` block:



4. Code the buttons

Now, you will code the arrow buttons to change the value of `repetitions` when the buttons are clicked.

Step 1:
Click on the **up** sprite. Leave the existing code as it is.
Select a `when this sprite clicked` block.

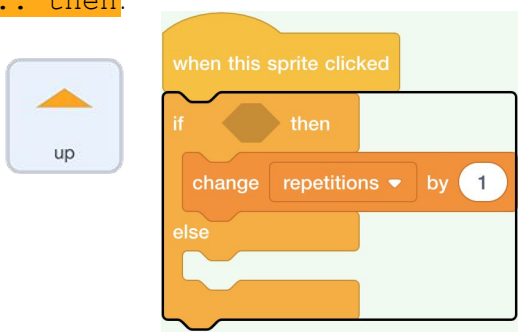


Step 2:
Add a `change repetitions` by block from the **`Variables`** blocks menu below. This should have a value of 1:



You don't want the exercise to repeat too many times, so next, you will add some code to stop `repetitions` from going above 5.

Step 3:
From the **`Control`** blocks menu, select an `if ... then ... else` block, and place the `change repetitions by 1` block in the space below `if ... then`:

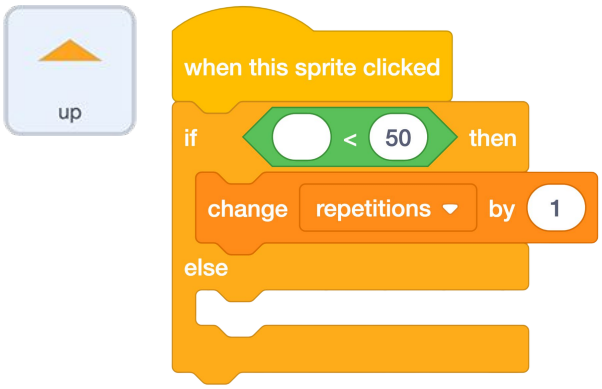


Now, you will add the condition `if repetitions is less than` 5.

Step 4:
From the **`Operators`** blocks menu, select a `less than` operator:



Step 5:
Place the block in the top space of the `if ... then ... else` block, between `if` and `then`.



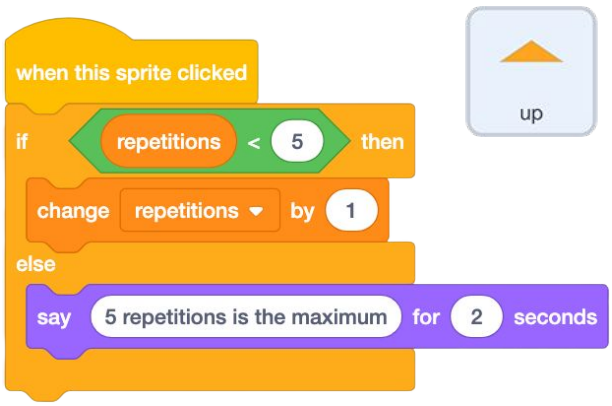
Now, `repetitions` should only go up by `1` `if` `repetitions` `is less than` `5`

Step 6:
From the `Variables` blocks menu, take a rounded `repetitions` block and place it before the `<`. Type the number `5` after the `<`.

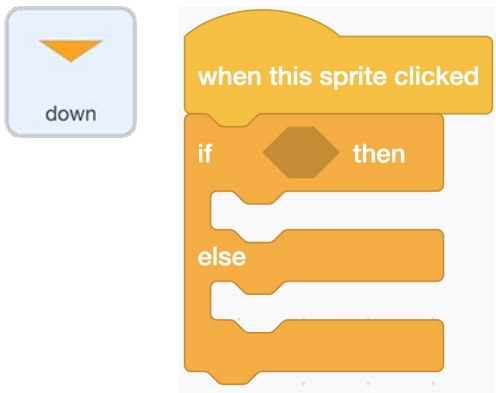


Now, you need to do the same for the **down** arrow, but you need to use a different condition and reduce `repetitions` by 1 each time. A way to do this in Scratch is `change by` `-1`.

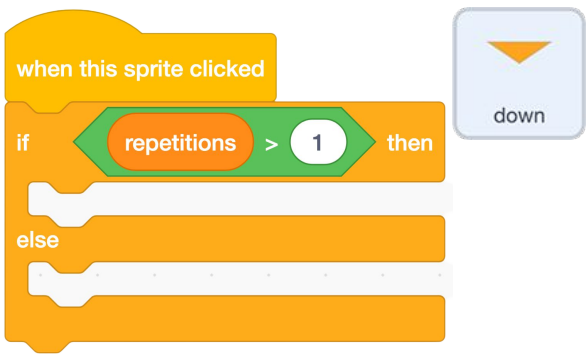
Step 7:
Below `else`, add a `say` block that will display the message `5 repetitions is the maximum` for `2` seconds:



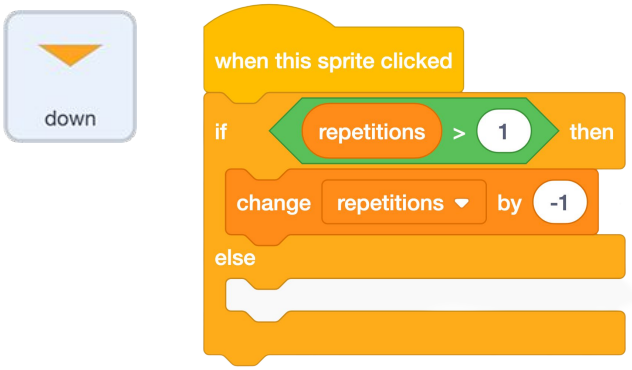
Step 8:
Select the **down** sprite. Add a `when this sprite clicked` block and an `if ... then ... else` block.



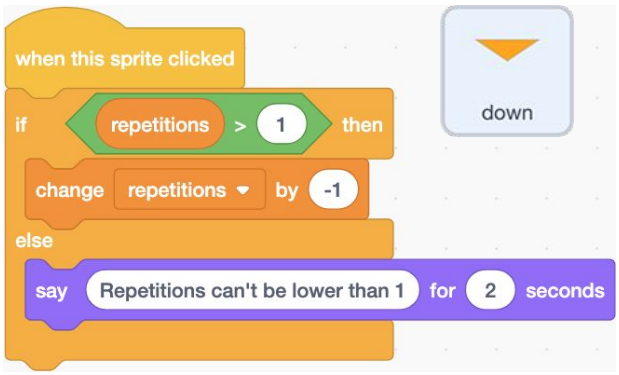
Step 9:
Add the condition `if` `repetitions` `is more than` `1`.



Step 10:
Add a block so that if the condition is **true**, your program will `change repetitions by` `-1`.



Step 11:
Add a block so that if the condition is **false**, your program will `say` `Repetitions can't be lower than 1` for `2` seconds:

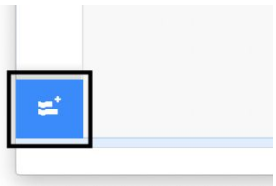


Step 12:
Run the program to test it. Click on the two arrows to increase or reduce the number of repetitions.

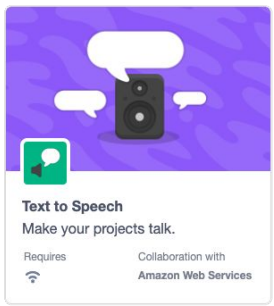
5. Add a head roll exercise

Now that you can control the number of times that each exercise repeats, it is time to add a second exercise routine.
For this exercise, you will use `Text to Speech` instead of `say` so that the exercise instructions will be spoken by the computer.

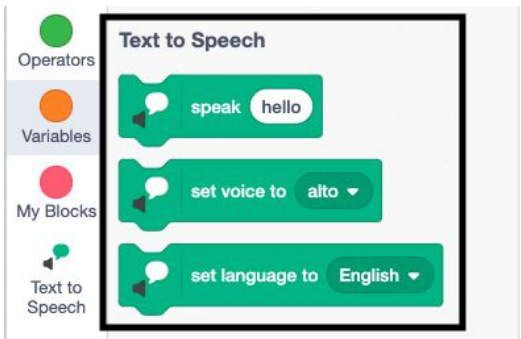
Step 1:
Click the **Add Extension** button in the bottom left-hand corner of the Scratch window.



Step 2:
Select the **Text to Speech** extension.

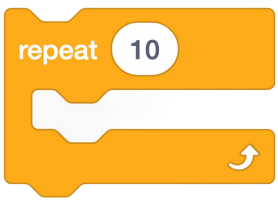


Step 3:
A new set of `Text to Speech` blocks will appear.

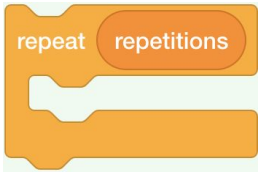


Step 4:
Choose the **Nadia** sprite and select a `repeat` block.



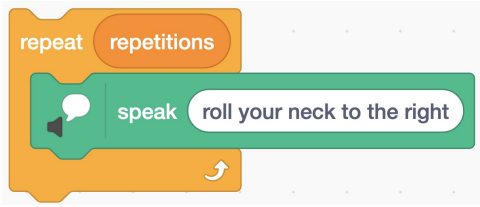


Step 3:
Add a rounded `repetitions` block from the `Variables` blocks menu to control the number of repetitions:

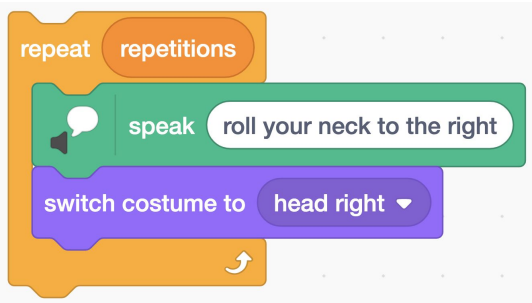


Now, you will start to add the exercise instructions.

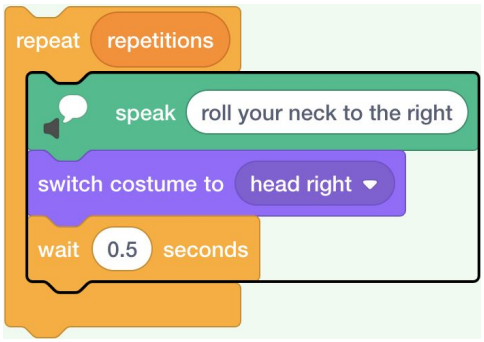
Step 4:
From the `Text to Speech` blocks menu, add a `speak` block and type in the text `roll your neck to the right`.



Step 5:
Next, add a `switch costume to` block and select the costume `head right`.

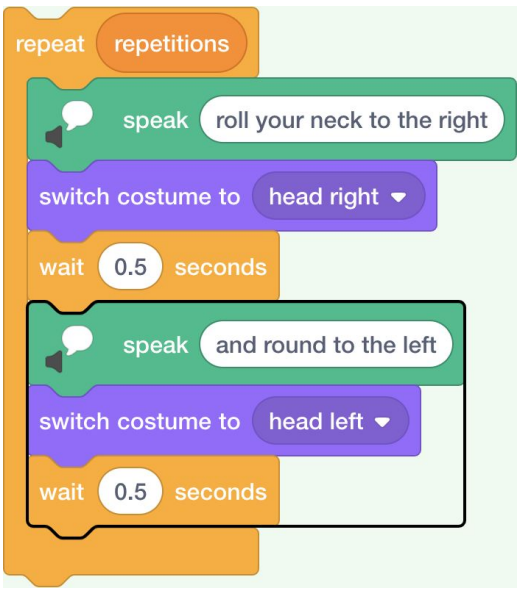


Step 6:
Finally, add a short `wait` block:

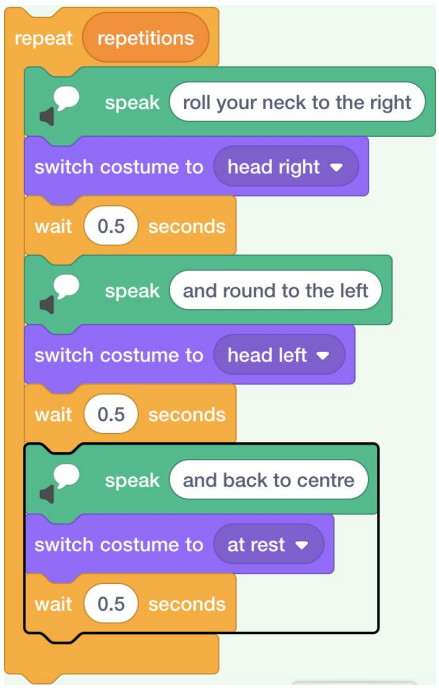


Step 7:

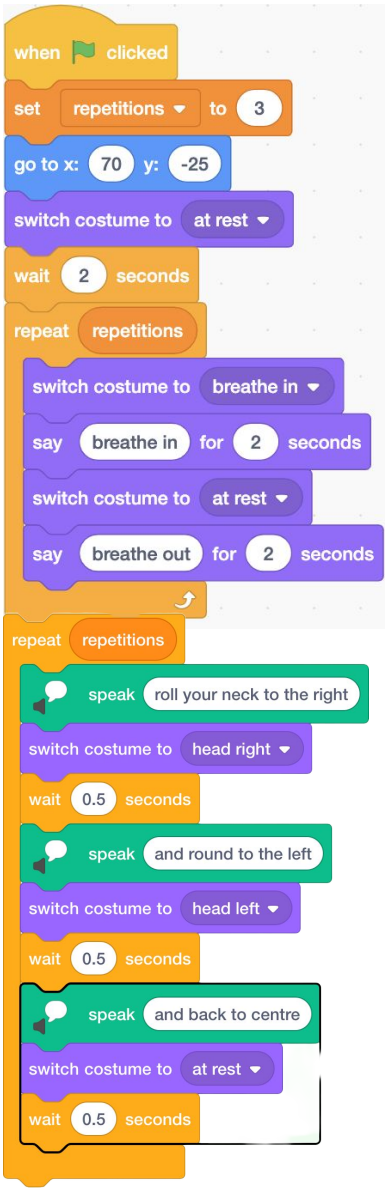
- Add the following `Text to Speech` instruction: `and round to the left`.
- Add blocks so that your **Nadia** sprite will `switch costume to` `head left` and `wait` for `0.5` seconds:



Step 8:
Type in the instruction `and back to centre`. Then, add a `switch costume to` block and set it back to `at rest`, then set it to `wait` for another `0.5` seconds:



Step 9:
Add your new exercise routine to your existing code by snapping the second `repeat` block under first `repeat`.



Save your project

6. Add more

Well done! You have programmed a healthy 'relax and stretch' exercise program to keep yourself and others moving.
You can now customise and improve your program for more exercises.

To create a new exercise routine you could use some of the **Nadia** sprite's costumes that are currently unused, or even duplicate an existing costume, and edit the costume to make a new position costume of your own!

It's completely up to you to use the `say` or `Text to Speech` block for the instructions.

To add controls that change the speed of the routines you could use a new `variable` to control the `wait` times and how long it takes to `say` the instructions. You could create more buttons to control speed, like the arrows to control the number of repetitions.

